AI531C — T12 · The Loss Ablation Studio: Student Sheet

Tue 3 Nov, 2:00–3:00 PM · run: python3 T12_studio.py (+ 'support' / 'extension')

Your mission

Watch three judges (MSE, MAE, SSIM) disagree about the same candidates, run one clean ablation, and assemble the six-row protocol for YOUR model.

Experiment

Judges' table: blur scores MSE 189 / SSIM 0.733; offset+16 scores MSE 255 / SSIM 0.951 — MSE prefers the BLUR (asserted). Then the ablation: patch count 5k vs 25k → 29.45 vs 29.48 dB (one variable, seeds fixed).

Vary / break it

The ablation's honest reading: 0.03 dB is NOISE-level — a finding of 'no effect' is still a finding. What variable would you ablate instead?

Discuss

Q1 · MSE preferred destroyed structure over a brightness offset. Derive who is 'right' for (a) radiology, (b) photo apps.

Q2 · Your ablation moved the metric by 0.03 dB. What TWO things must you check before claiming 'no effect'?

Q3 · Why must the ablation hold seeds fixed — what does seed variation masquerade as?

Reflect

The protocol table you filled today is the capstone's evaluation section, verbatim (L34's promise).

Fast? EXTENSION cell: sweep the mix loss $\lambda \in \{0, 0.2, 0.5\}$; find where the blur/offset preference flips. Struggling? SUPPORT cell: judges' table only + the medical-danger question. Nothing is due after class — the hour is the work.